TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	72.7 ± 1.25
Correlation Stability	87.0 ± 1.41
Distribution Consistency	84.3 ± 3.68
Entropy Preservation	82.0 ± 2.16
Mean Z Score Drift	0.3 ± 0.07
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	32.7 ± 7.36
Data Density	99.1 ± 0.31
Coverage Score	42.3 ± 11.26
Feature Accuracy	79.3 ± 6.13
Novelty Score	0.4 ± 0.12

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Dataset Audit: parkinsons.csv

1. Dataset Portrait

Attribute	Value
Rows	195
Columns	23
Numeric features	23
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	3
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
MDVP:Fo	154.229	156.348	+2.119	1.4%	OK
MDVP:Fhi	197.105	190.098	-7.007	3.6%	OK
MDVP:Flo	116.325	110.157	-6.168	5.3%	OK
MDVP:Jitter	0.006	0.007	+0.001	11.4%	Warn
MDVP:Jitter.1	0.006	0.007	+0.001	11.4%	Warn
MDVP:RAP	0.003	0.004	+0.000	8.3%	OK
MDVP:PPQ	0.003	0.004	+0.000	9.9%	OK
Jitter:DDP	0.010	0.010	+0.000	2.9%	OK
MDVP:Shimmer	0.030	0.032	+0.002	8.4%	OK
MDVP:Shimmer.1	0.030	0.032	+0.002	8.4%	OK
Shimmer:APQ3	0.016	0.016	+0.001	4.3%	OK
Shimmer:APQ5	0.018	0.019	+0.001	6.6%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	72.7 ± 1.25
Correlation Stability (*2)	87.0 ± 1.41
Distribution Consistency (*3)	84.3 ± 3.68
Entropy Preservation (*4)	82.0 ± 2.16
Mean Z Score Drift (*5)	0.3 ± 0.07
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	32.7 ± 7.36

Data Density (*8)	99.1 ± 0.31
Coverage Score (*9)	42.3 ± 11.26
Feature Accuracy (*10)	79.3 ± 6.13
Novelty Score (*11)	0.4 ± 0.12

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9743589743589743	0.8974358974358975	0.9487179487179486	-0.02564102564102563	
F1	0.9806435394670696	0.9230719661820806	0.9622806387512269	-0.0183629007158419	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8803418803418804	0.8888888888888888	0.8461538461538461	-0.03418803418803418	
F1	0.9096765824974057	0.9181231466945752	0.8833268858800772	-0.026349696617328406	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9658119658119658	0.8119658119658121	0.9316239316239315	-0.03418803418803418	
F1	0.9748397136632431	0.8613553491572067	0.9497601323407775	-0.0250795813224655	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9487179487179486	0.8803418803418803	0.9658119658119658	0.01709401709401707	
F1	0.9609674981103553	0.9130864197530865	0.9738461538461539	0.012878655735798605	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7777777777777778	0.7094017094017094	0.7948717948717948	0.01709401709401710 6	
F1	0.8614267676767677	0.8125486305633364	0.8674096736596737	0.005982905982905962	

Model: LR

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8803418803418803	0.8290598290598291	0.8632478632478633	-0.01709401709401707	
F1	0.91232423490488	0.8857605720350819	0.90339064266958	-0.008933592235300067	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8461538461538463	0.717948717948718	0.8547008547008547	0.008547008547008517	
F1	0.8918089248277927	0.8041050091577754	0.8959511077158137	0.004142182888020818	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.717948717948718	0.5299145299145299	0.6239316239316239	-0.09401709401709402	
F1	0.749906191369606	0.49728622631848435	0.628552784998342	-0.12135340637126406	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7350427350427351	0.4615384615384615	0.8034188034188036	0.06837606837606836	
F1	0.8266002415458938	0.4947414167753151	0.8685628821421041	0.04196264059621043	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9401709401709403	0.8888888888888888	0.8888888888888889	-0.05128205128205129	
F1	0.9570703384745131	0.9172113289760349	0.9190973360176038	-0.03797300245690923	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	119.992	157.302	74.997	0.00784	0.00784	0.0037	0.00554	0.01109	0.04374	0.04374
2	122.4	148.65	113.819	0.00968	0.00968	0.00465	0.00696	0.01394	0.06134	0.06134
3	116.682	131.111	111.555	0.0105	0.0105	0.00544	0.00781	0.01633	0.05233	0.05233
4	116.676	137.871	111.366	0.00997	0.00997	0.00502	0.00698	0.01505	0.05492	0.05492
5	116.014	141.781	110.655	0.01284	0.01284	0.00655	0.00908	0.01966	0.06425	0.06425
6	120.552	131.162	113.787	0.00968	0.00968	0.00463	0.0075	0.01388	0.04701	0.04701
7	120.267	137.244	114.82	0.00333	0.00333	0.00155	0.00202	0.00466	0.01608	0.01608
8	107.332	113.84	104.315	0.0029	0.0029	0.00144	0.00182	0.00431	0.01567	0.01567
9	95.73	132.068	91.754	0.00551	0.00551	0.00293	0.00332	0.0088	0.02093	0.02093
10	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
11	88.333	112.24	84.072	0.00505	0.00505	0.00254	0.0033	0.00763	0.02143	0.02143
12	91.904	115.871	86.292	0.0054	0.0054	0.00281	0.00336	0.00844	0.02752	0.02752
13	136.926	159.866	131.276	0.00293	0.00293	0.00118	0.00153	0.00355	0.01259	0.01259
14	139.173	179.139	76.556	0.0039	0.0039	0.00165	0.00208	0.00496	0.01642	0.01642
15	152.845	163.305	75.836	0.00294	0.00294	0.00121	0.00149	0.00364	0.01828	0.01828
16	142.167	217.455	83.159	0.00369	0.00369	0.00157	0.00203	0.00471	0.01503	0.01503
17	144.188	349.259	82.764	0.00544	0.00544	0.00211	0.00292	0.00632	0.02047	0.02047
18	168.778	232.181	75.603	0.00718	0.00718	0.00284	0.00387	0.00853	0.03327	0.03327
19	153.046	175.829	68.623	0.00742	0.00742	0.00364	0.00432	0.01092	0.05517	0.05517
20	156.405	189.398	142.822	0.00768	0.00768	0.00372	0.00399	0.01116	0.03995	0.03995

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	157.821	172.975	68.401	0.00358	0.00358	0.00196	0.00196	0.00587	0.03716	0.03716
2	171.041	208.313	75.501	0.00455	0.00455	0.0025	0.00234	0.0075	0.01966	0.01966
3	198.116	233.099	174.478	0.02714	0.02714	0.01568	0.01699	0.04705	0.08684	0.08684
4	177.876	192.921	168.013	0.00411	0.00411	0.00233	0.00241	0.007	0.02126	0.02126
5	117.963	134.209	100.757	0.01813	0.01813	0.01117	0.00718	0.03351	0.04912	0.04912
6	202.266	211.604	197.079	0.0018	0.0018	0.00093	0.00107	0.00278	0.00954	0.00954
7	150.123	180.456	70.789	0.00412	0.00412	0.00201	0.00205	0.00645	0.03812	0.03812
8	165.789	200.123	80.456	0.00567	0.00567	0.00267	0.00245	0.00789	0.02045	0.02045
9	190.567	220.789	175.123	0.02845	0.02845	0.01678	0.01789	0.04812	0.08845	0.08845
10	180.123	195.456	170.789	0.00456	0.00456	0.00245	0.00256	0.00745	0.02234	0.02234
11	120.456	135.789	102.123	0.01945	0.01945	0.01123	0.00789	0.03456	0.05012	0.05012
12	205.123	215.456	200.789	0.00234	0.00234	0.00123	0.00145	0.00345	0.01012	0.01012
13	110.417	131.067	103.37	0.00784	0.00784	0.00352	0.00514	0.01056	0.03715	0.03715
14	208.083	253.792	91.802	0.00757	0.00757	0.00428	0.00428	0.01285	0.06725	0.06725
15	244.99	272.21	239.17	0.00451	0.00451	0.00279	0.00237	0.00837	0.01897	0.01897
16	228.969	239.541	113.201	0.00238	0.00238	0.00136	0.0014	0.00408	0.01745	0.01745

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	156.239	195.107	79.82	0.00694	0.00694	0.00403	0.00395	0.01209	0.02343	0.02343
2	192.818	224.429	168.793	0.03107	0.03107	0.018	0.01958	0.05401	0.11908	0.11908
3	237.226	247.326	225.227	0.00298	0.00298	0.00169	0.00182	0.00507	0.01752	0.01752
4	176.858	205.56	75.344	0.00747	0.00747	0.00418	0.0042	0.01254	0.04795	0.04795
5	117.004	144.466	99.923	0.00353	0.00353	0.00176	0.00218	0.00528	0.01657	0.01657
6	200.714	223.982	89.686	0.00621	0.00621	0.00371	0.00337	0.01112	0.02286	0.02286
7	155.432	198.765	80.123	0.00712	0.00712	0.00456	0.00434	0.01245	0.02456	0.02456
8	191.234	220.456	170.123	0.03212	0.03212	0.01934	0.02045	0.05567	0.12034	0.12034
9	238.123	250.456	226.789	0.00345	0.00345	0.00178	0.00189	0.00567	0.0189	0.0189
10	178.123	207.89	76.456	0.00812	0.00812	0.00456	0.00445	0.01345	0.0489	0.0489
11	119.123	146.789	100.123	0.00378	0.00378	0.00189	0.00245	0.00567	0.01789	0.01789
12	201.123	225.678	90.123	0.00678	0.00678	0.00345	0.00312	0.01123	0.02345	0.02345
13	245.51	262.09	231.848	0.00235	0.00235	0.00127	0.00148	0.0038	0.01608	0.01608
14	115.38	123.109	108.634	0.00332	0.00332	0.0016	0.00199	0.0048	0.01503	0.01503
15	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
16	143.533	162.215	65.809	0.01101	0.01101	0.00647	0.00467	0.01941	0.05384	0.05384

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	152.125	161.469	76.596	0.00382	0.00382	0.00191	0.00226	0.00574	0.05925	0.05925
2	193.03	208.9	80.297	0.00766	0.00766	0.0045	0.00389	0.01351	0.03044	0.03044
3	188.62	216.302	165.982	0.01719	0.01719	0.00996	0.01027	0.02987	0.07118	0.07118
4	176.17	185.604	163.564	0.00369	0.00369	0.00205	0.00218	0.00616	0.01851	0.01851
5	108.807	134.656	102.874	0.00761	0.00761	0.00349	0.00486	0.01046	0.02719	0.02719
6	122.964	130.049	114.676	0.00428	0.00428	0.00124	0.00155	0.00373	0.01681	0.01681
7	150.5	160.2	75.3	0.0045	0.0045	0.002	0.0025	0.006	0.058	0.058
8	190.0	205.0	78.0	0.008	0.008	0.004	0.0035	0.012	0.029	0.029
9	185.0	220.0	160.0	0.016	0.016	0.01	0.01	0.03	0.07	0.07
10	175.0	190.0	160.0	0.0035	0.0035	0.002	0.002	0.005	0.02	0.02
11	110.0	135.0	100.0	0.007	0.007	0.003	0.005	0.01	0.028	0.028
12	125.0	130.0	115.0	0.004	0.004	0.0015	0.0015	0.0035	0.017	0.017
13	242.852	255.034	227.911	0.00225	0.00225	0.00117	0.00139	0.0035	0.01494	0.01494
14	139.224	586.567	66.157	0.03011	0.03011	0.01854	0.01628	0.05563	0.09419	0.09419
15	186.163	197.724	177.584	0.00298	0.00298	0.00165	0.00175	0.00496	0.01495	0.01495
16	116.879	131.897	108.153	0.00788	0.00788	0.00334	0.00493	0.01003	0.02645	0.02645